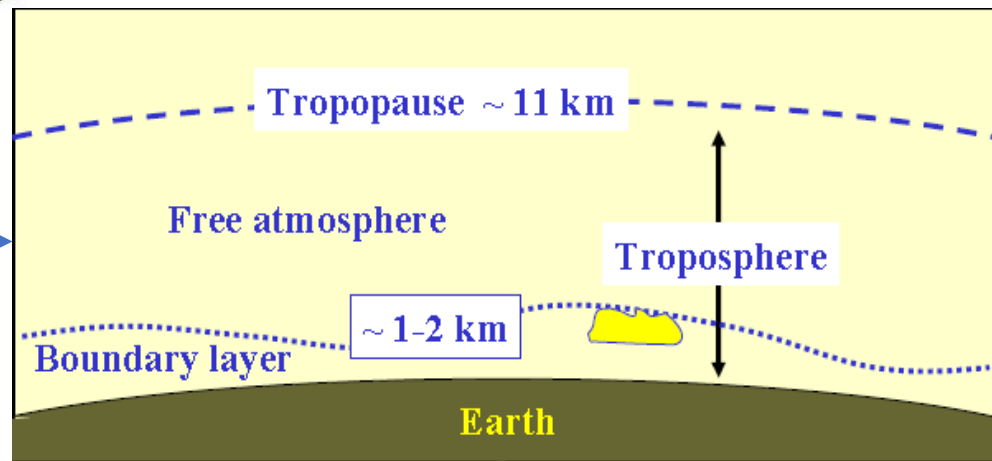
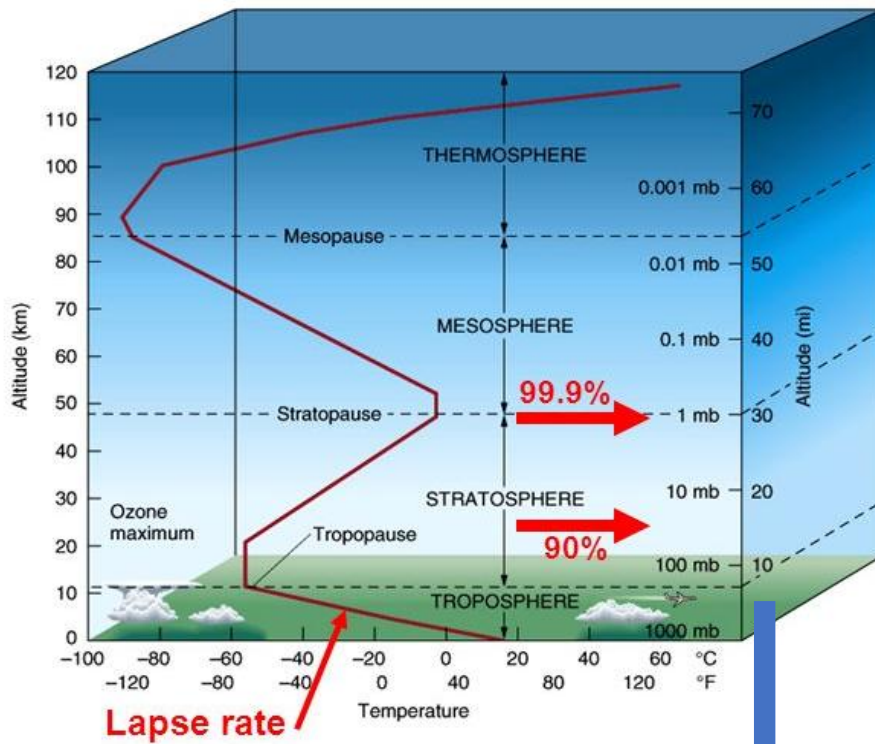




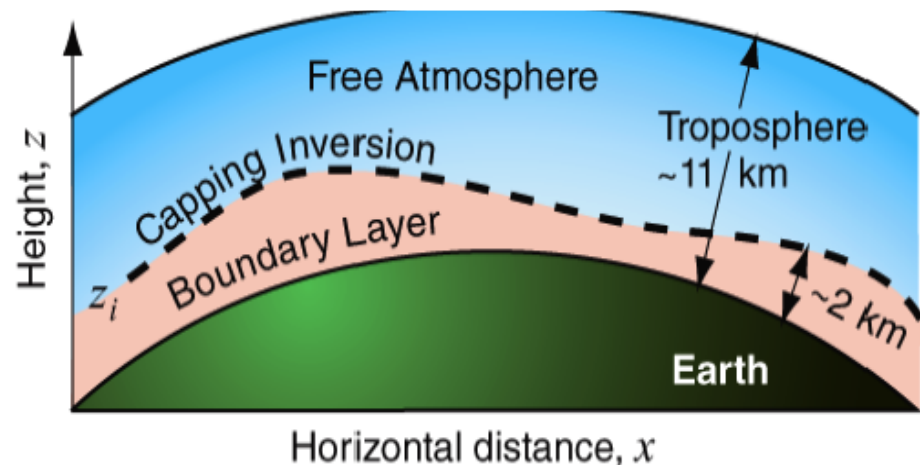
Atmospheric Boundary Layer

Atmospheric Boundary Layer



ABL: Basic Properties

- It is the lowest part of tropospheric layer that is **directly influenced by the Earth's surface** and **responds to surface forcings**. It is also called the planetary boundary layer (PBL) or just the boundary layer (BL).
- This phenomena in general **very short time scales** which covers from few hours (maximum happens within the day).
- It forms as a result of turbulent exchange process between surface and the atmosphere.
- Basically, this layer is **turbulent** and is **well mixed**. Turbulence is generated by wind shear.
- Typical height of the ABL: **usually ~1 - 2 km** but can vary
- ~3 - 5 km over deserts, dry fields and forests

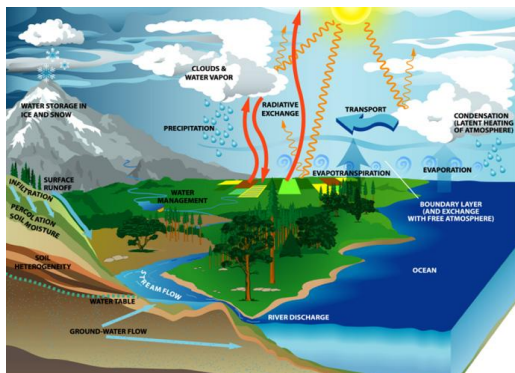


Kinds of atmospheric boundary layers

Marine	Continental
Little diurnal variability	Strong diurnal variability
1-2 km	Up to 5 km over deserts
Low Bowen ratio	High Bowen ratio
Wave state important	Surface shape fixed, but important

Why studying the ABL/PBL?

- Humans live in the ABL – “Natural environment for human activities”
- Agriculture, aeronautics, telecommunications
- Weather and climate prediction, pollutants dispersion
- ABL is the kind of Fuel layer for most of the catastrophic weather events such as Cyclones, Thunderstorms, Fog formation... etc
- It is a reservoir of trace gases and pollutants.
- Pollution traps in the ABL (can be viewed as a pollution layer)



Things that happen in the ABL

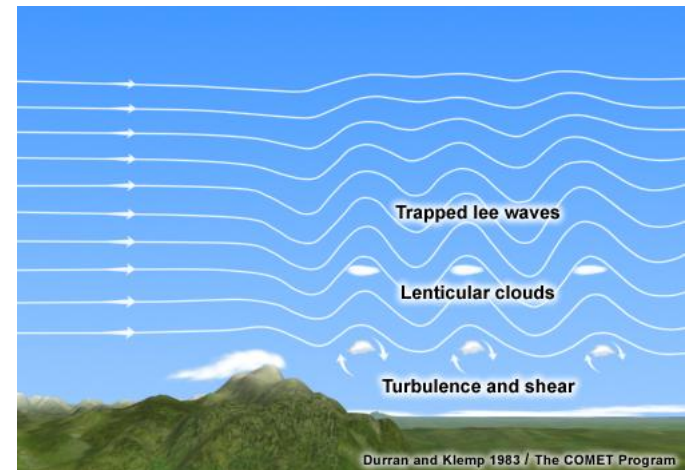
- Large diurnal temperature variations (relative to the free troposphere above).
- Turbulence

Sources of turbulence:

Thermal forcing (thermals: buoyant eddies forced by solar heating of the surface)

Vertical wind shear (due to frictional drag by the surface on flow)

- Horizontal wind shear (due to flow of wind around obstacles: trees, mountains, islands)
- Waves (gravity)
- Spiraling Winds
- Clouds
- Fog





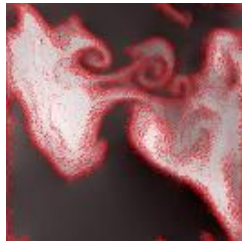
Synoptic and mesoscale systems

Free Atmosphere

- vertical transport
by advection



Waves



Turbulence



Convection

Boundary layer

-vertical transport
by turbulent mixing

-communicates
influence of
underlying surface
properties to the
free atmosphere



Factors influencing the ABL

- The structure of the ABL is determined by the properties of TURBULENCE within it. Turbulence can be generated mechanically (through shear flow instability) or thermally (through convection in an unstably-stratified flow).

Shear

Surface
properties

Roughness
Orography

Buoyancy

Radiation (LW & SW)

Albedo

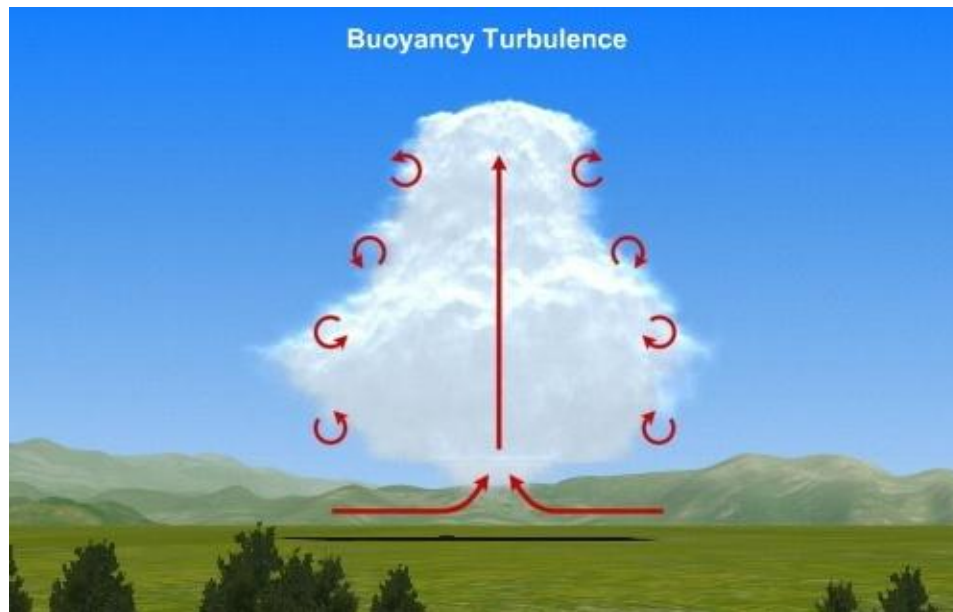
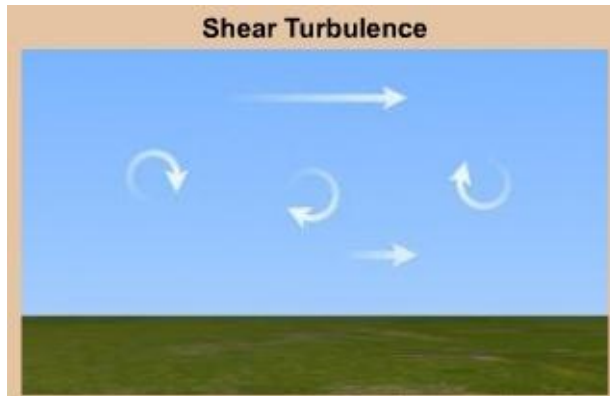
Emissivity

Sfc. Temperature

Wetness

What is turbulence?

Turbulence can be visualized as consisting of irregular swirls of motion called eddies. Eddies can vary in size, however, eddies exist in size as small as a few millimeters



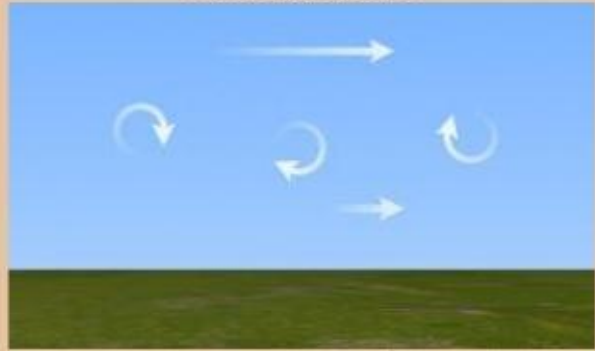
Eddies transport heat, moisture and momentum. The mechanical impact of all sub grid-scale motions upon an observable flow is collectively called a frictional force.

Thermal turbulence is caused by solar heating of the surface, which in turn heats the lower atmosphere resulting in uneven convective currents, which lead to turbulence.

What creates turbulence?

- *Solar heating generating thermals - nothing more than larger eddies (irregular swirls of motion)*
- *wind shear*
- *deflected flow around obstacles such as trees and buildings, creating turbulent wakes downstream of the obstacle*
 - *In summary, turbulence allows the boundary layer to respond to changes in surface forcings (e.g., daytime heating).*

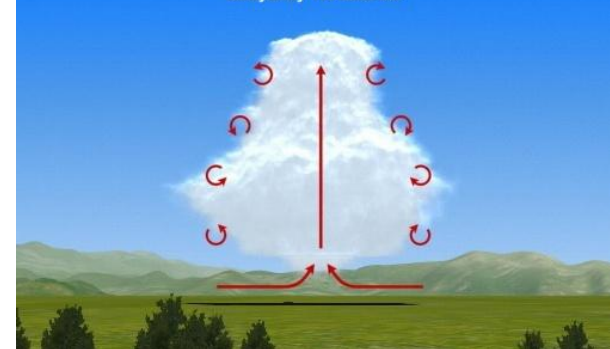
Shear Turbulence



Mechanical Turbulence



Buoyancy Turbulence

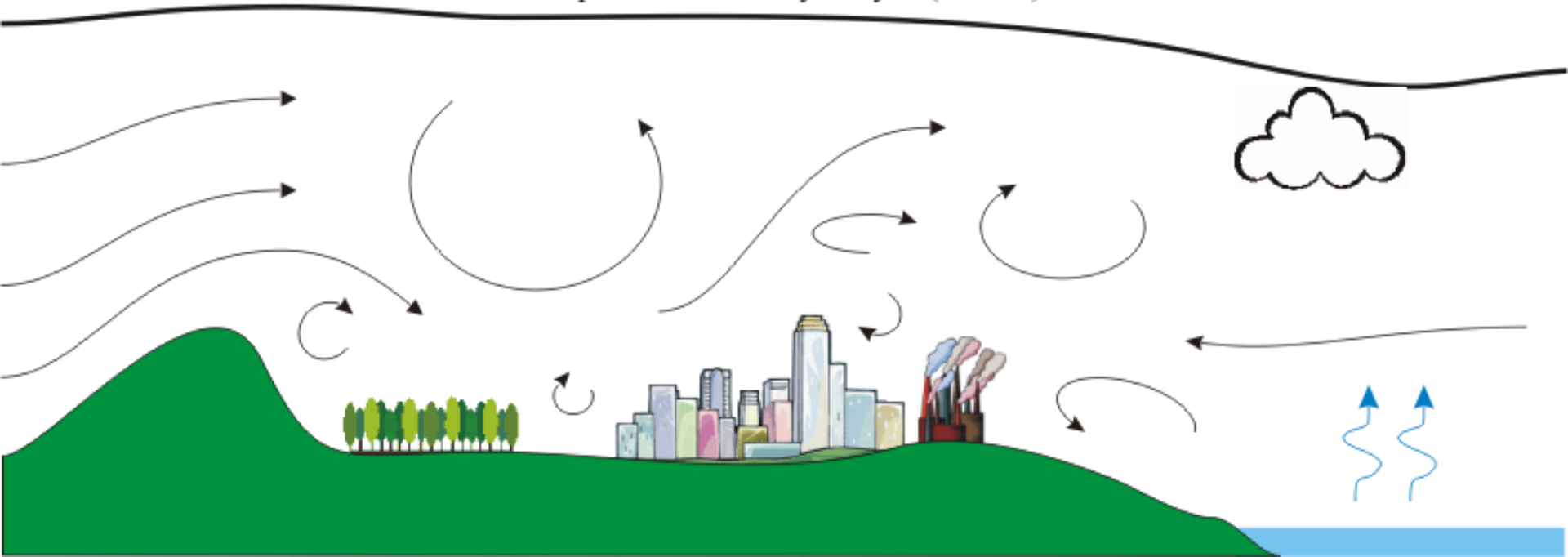


This does not occur in the free troposphere, the free troposphere acts like as if the earth's surface does not exist.

Boundary layer turbulence

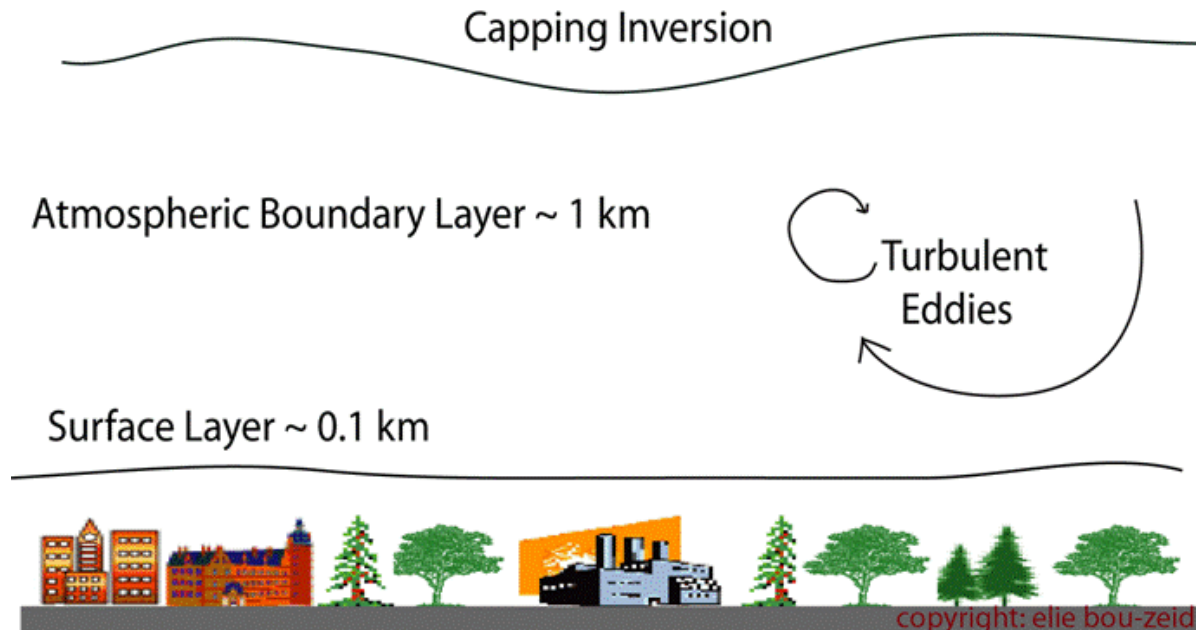
- When vertical wind shear is small and there is no buoyancy, the flow is laminar, or non-turbulent.
- For larger vertical wind shear, the flow becomes mechanically turbulent. - **Predominant during the nighttime hours.**
- In the presence of buoyancy, the flow becomes convectively turbulent - **Dominant during the daytime hours.**

Atmospheric Boundary Layer (~ 1km)



What goes on in the boundary layer

- Heat and moisture are mixed upward from the surface, where they are extracted via surface fluxes.
- Momentum is typically mixed downward in an environment with wind speed increasing with height.



Richardson number

In a statically stable layer, turbulence can exist only if mechanical production is large enough to overcome the damping effects of stability and viscous dissipation.

This condition is measured by a quantity called the **flux Richardson number**.

It is defined as

$$R_f = \frac{\frac{g}{\bar{\theta}} \overline{w'\theta'}}{\overline{u'w'} \frac{\partial \bar{u}}{\partial z} + \overline{v'w'} \frac{\partial \bar{v}}{\partial z}} = \frac{\text{buoyancy}}{\text{mechanical shear}}$$
$$R_f < 1 \quad (> 1) \Rightarrow \text{Flow is turbulent (laminar).}$$

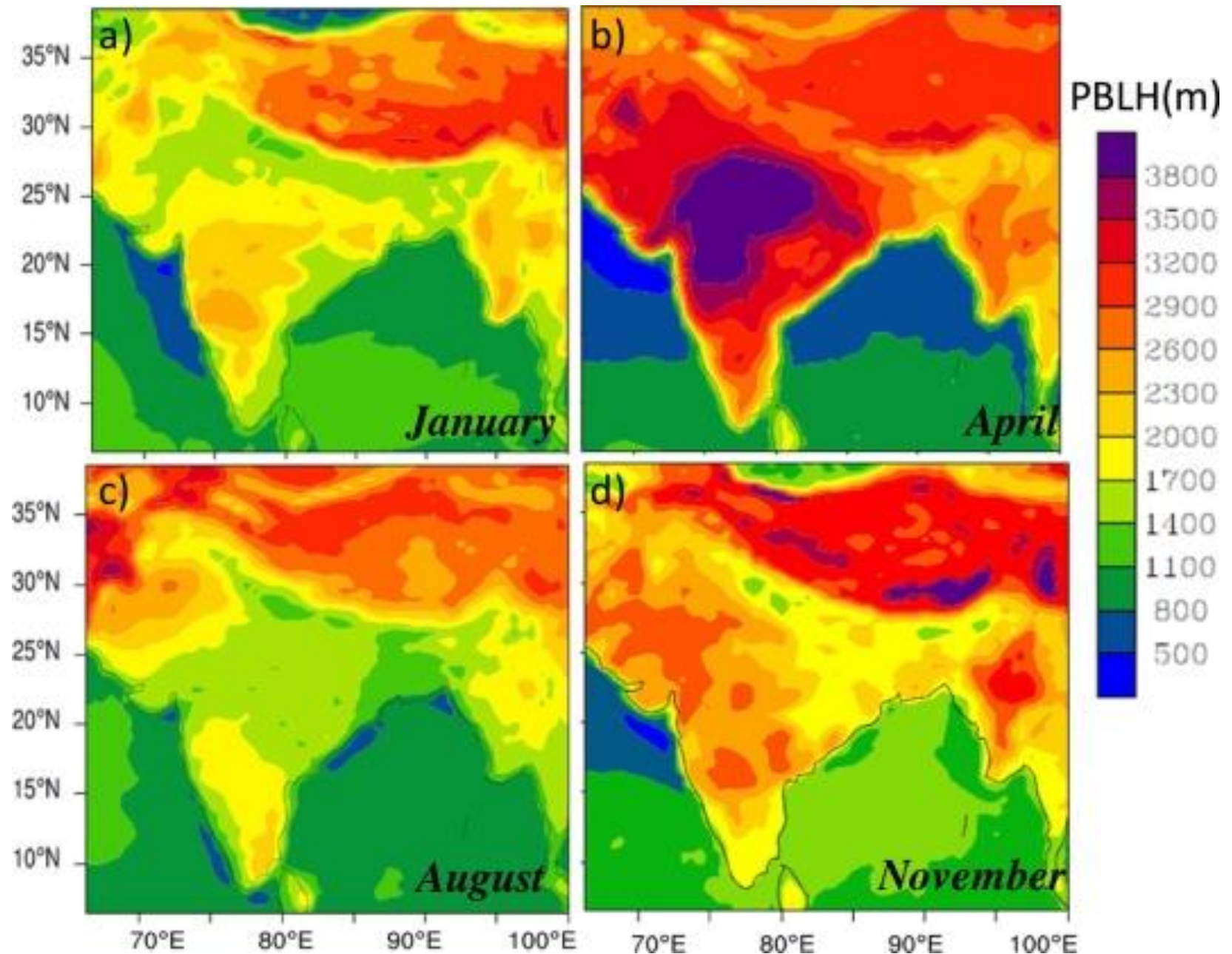
Following eddy diffusivity theory, alternatively, this can be defined using

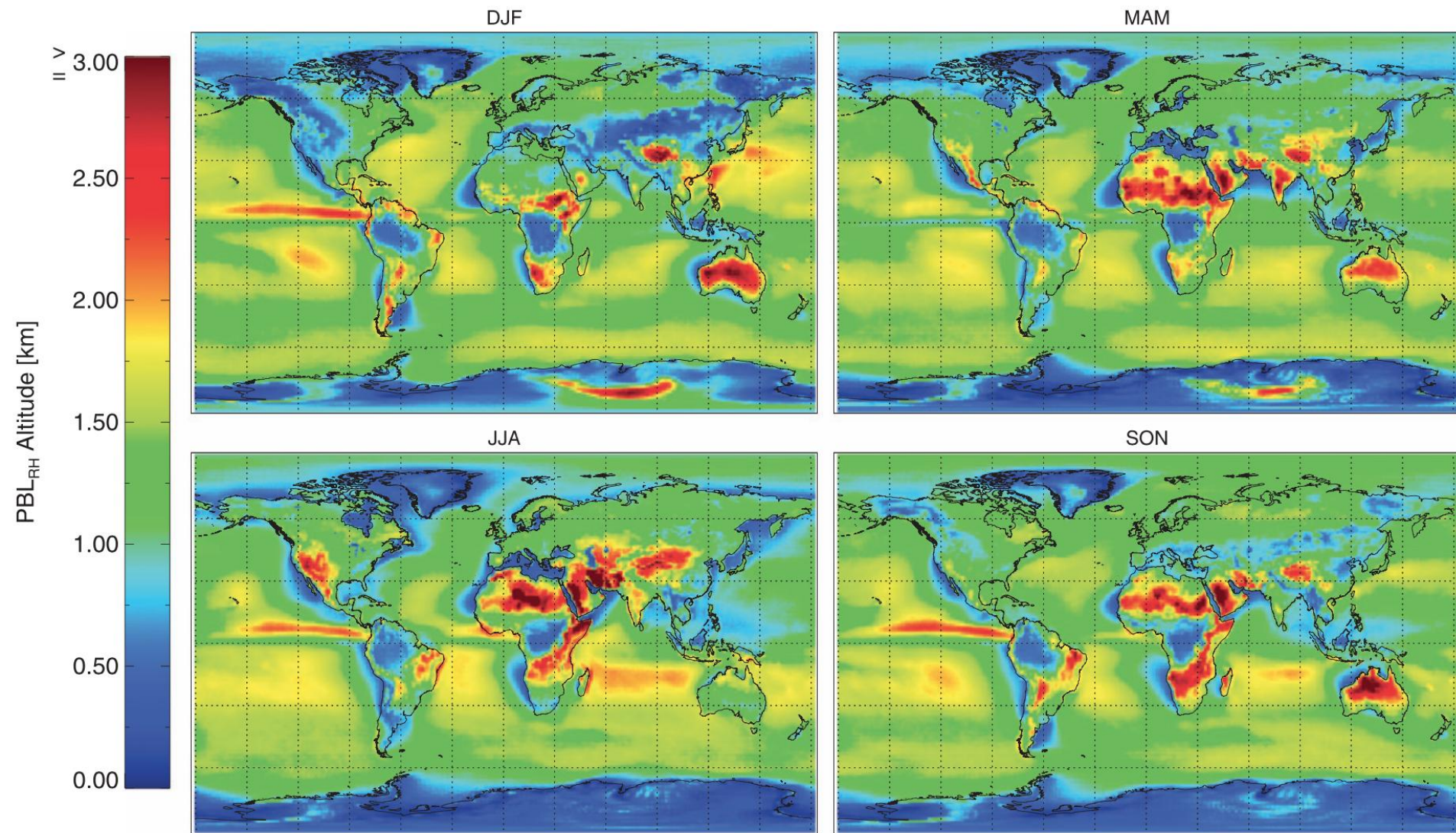
Richardson number (Wallace and Hobbs 2006)

$$Ri = \frac{g}{\bar{\theta}} \frac{\partial \bar{\theta}}{\partial z} \bigg/ \left[\left(\frac{\partial \bar{u}}{\partial z} \right)^2 + \left(\frac{\partial \bar{v}}{\partial z} \right)^2 \right] \quad Ri < \frac{1}{4} \quad \text{Laminar flow becomes turbulent}$$

(For more details, read Kelvin-Helmholtz instability theory)

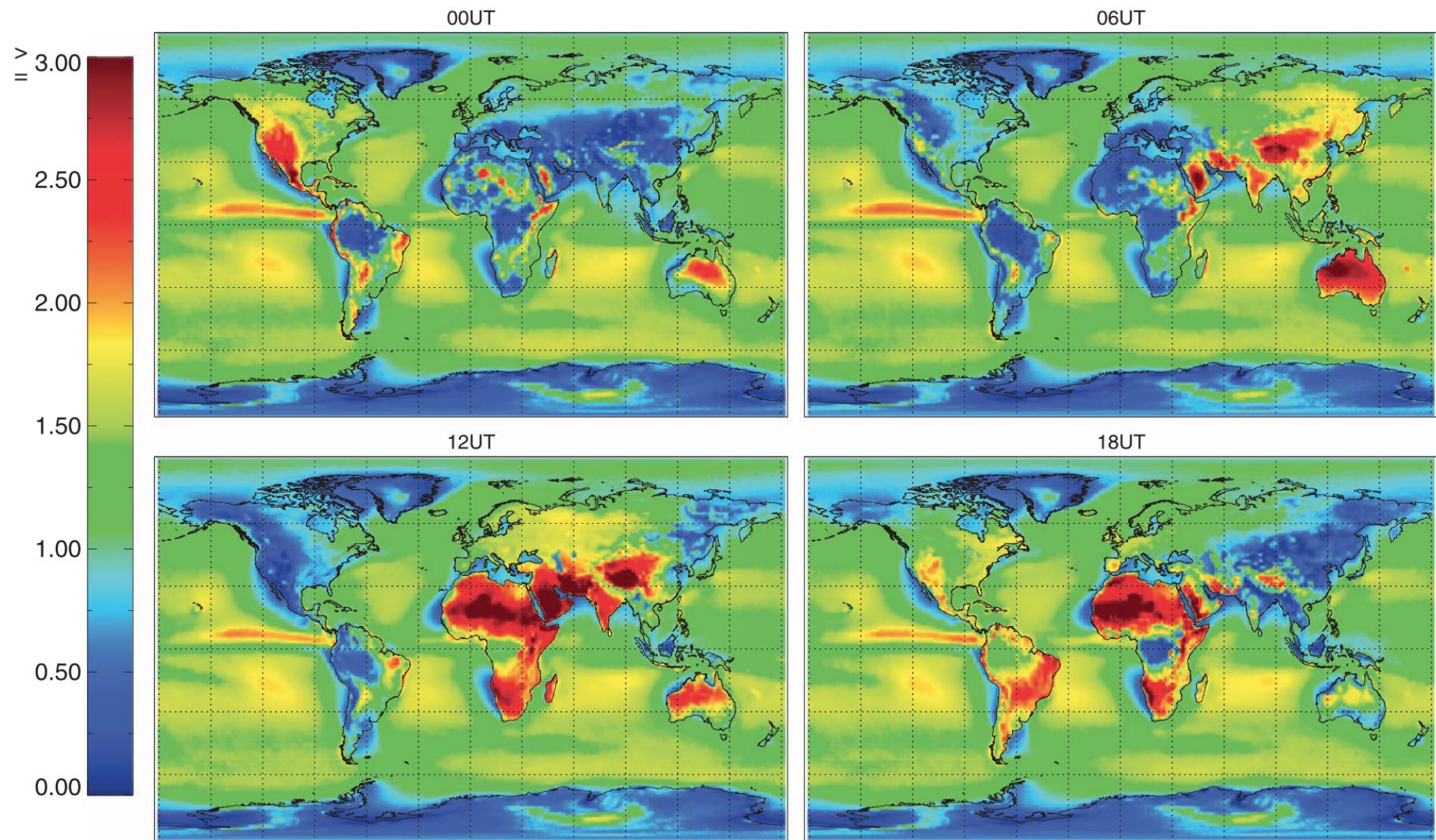
Seasonal Variations in the PBL height over India





Engeln et al. 2013

- PBLH is higher in Tropical region compared to Polar region
- Deep PBLH is to be noted during the Pre-Summer & Low in Winter
- Weaker Significant seasonal cycle over the Oceans compared to land

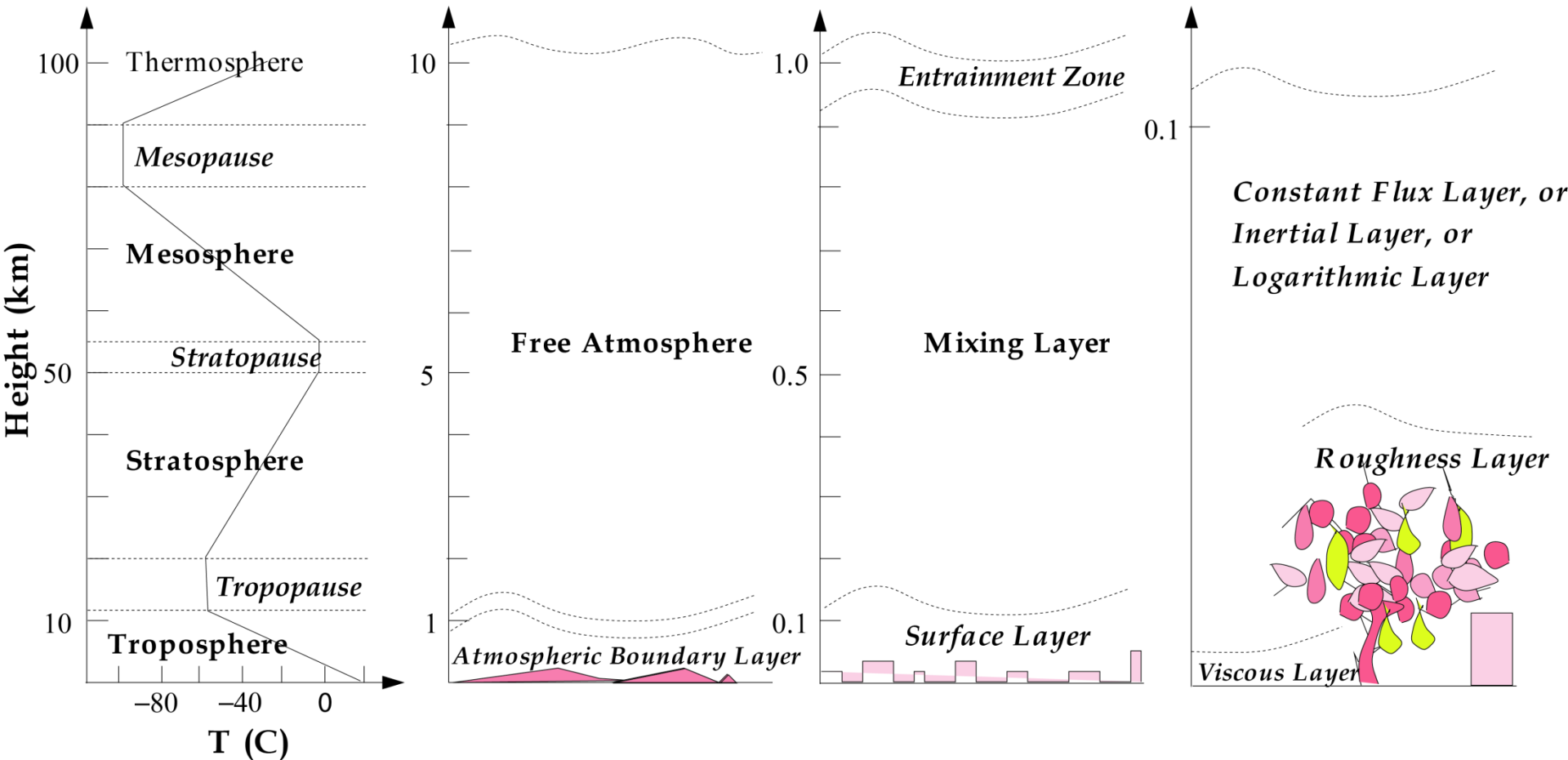


- No discernible diurnal cycle **over the ocean** & but **over land**,
- Diurnal cycle is intense, with large values of PBLH height over **Asia and Australia** at 0600 UTC, **Africa and the Middle East** at 1200 UTC, and **the Americas** at 0000 and 1800 UTC.
- These large values are found at very similar local times of around **12 to 17 hours**.

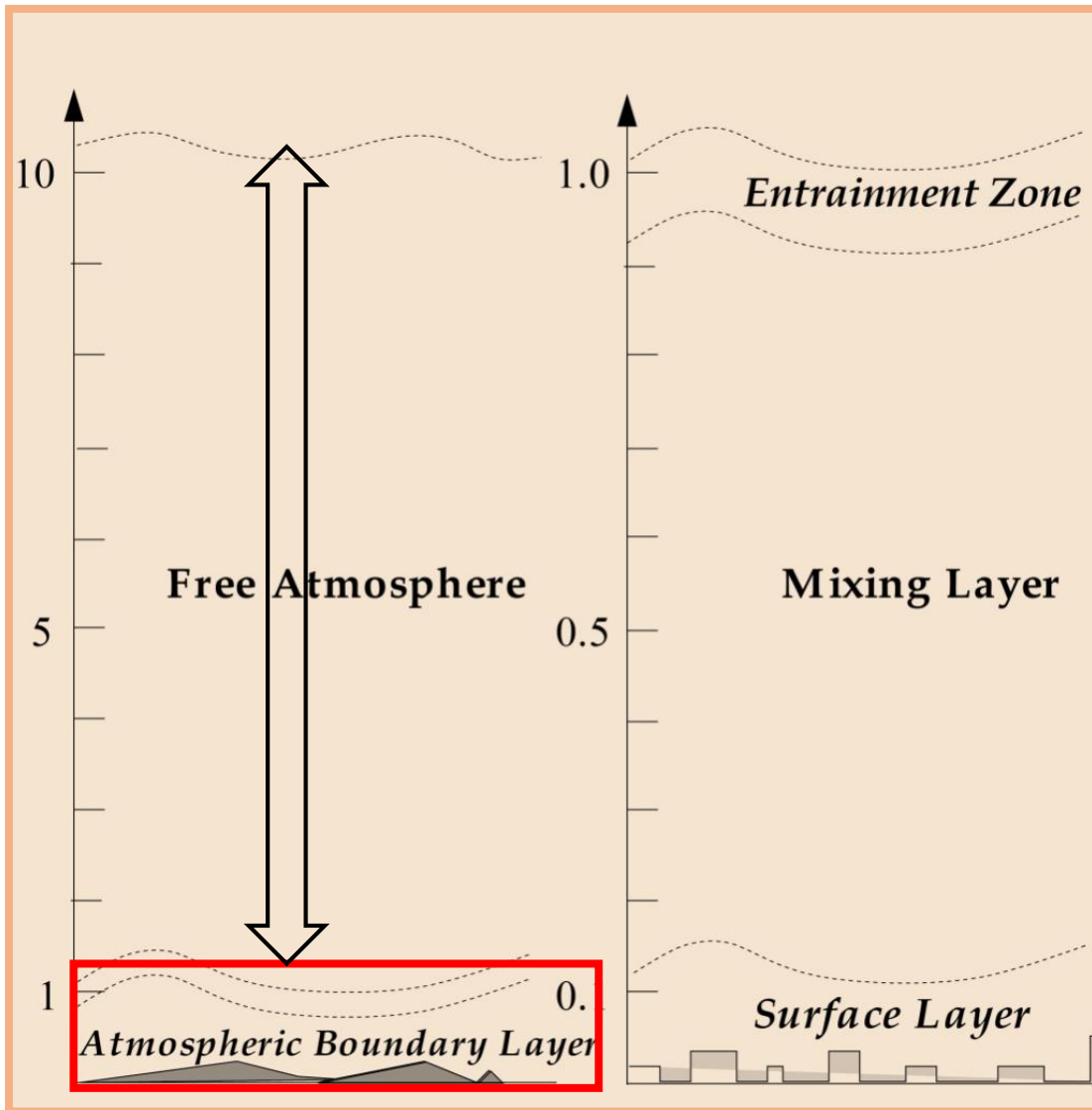
Scales of horizontal motion in the atmosphere

Larger than	Scale	Name
20,000 km		Planetary scale
2,000 km		Synoptic scale
200 km	Meso- α	} Mesoscale
20 km	Meso- β	
2 km	Meso- γ	
200 m	Micro- α	Boundary-layer turbulence
20 m	Micro- β	Surface-layer turbulence
2 m	Micro- γ	Inertial subrange turbulence
2 mm	Micro- δ	Fine-scale turbulence
Air molecules	Molecular	Viscous dissipation subrange

Layers of ABL



Vertical Structure of the ABL



PBL roughly divide into three layers:

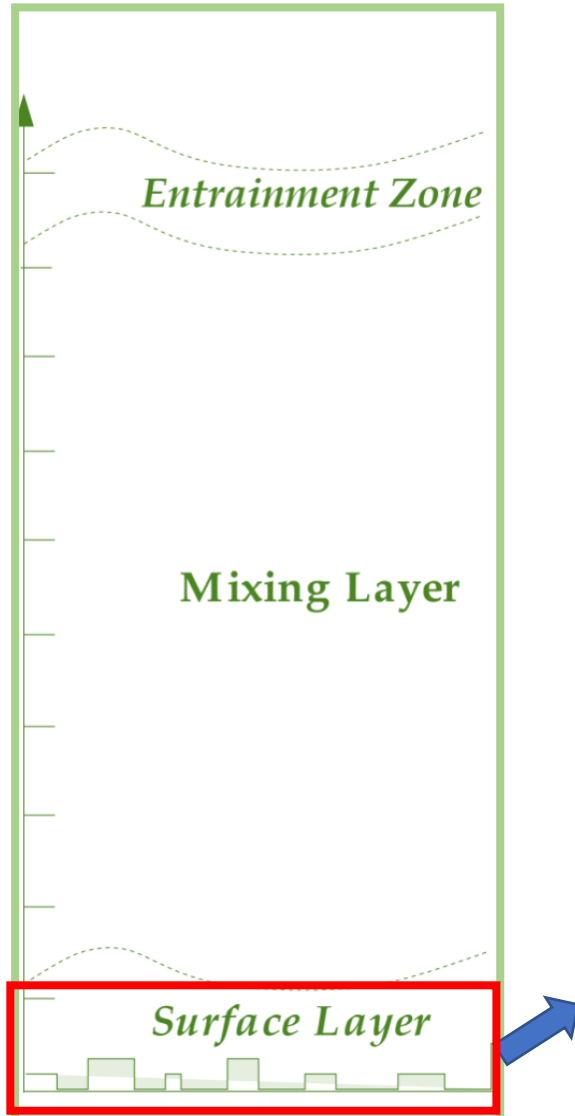
Surface Layer (SL)
Mixed Layer (ML)
Entrainment Zone (EZ)

Stable Boundary Layer (SBL)/Nocturnal BL
Residual Layer (RL)
Capping Inversion (CI)

ABL roughly divide into three regions:

1. The surface layer

The layer where the mechanical (shear) generation of turbulence exceeds the thermal turbulence (buoyant)



The layer occupies lowest 10% of the depth of ABL

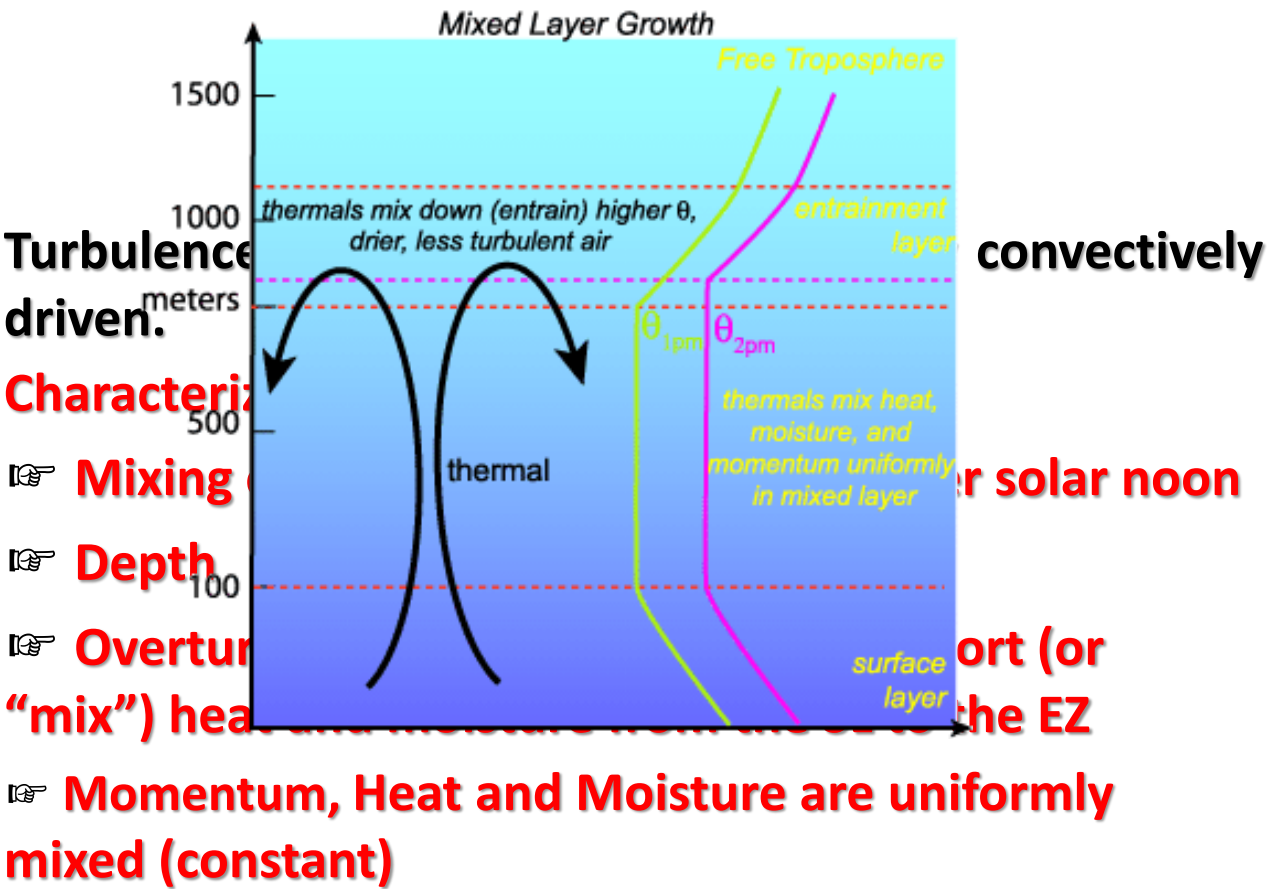
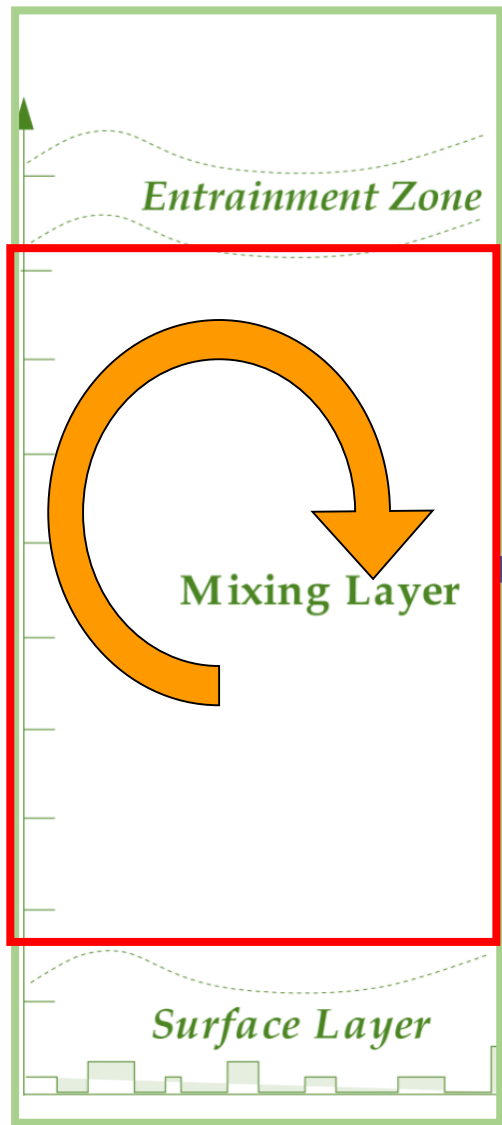
Characterized by:

- ☞ **Strong vertical gradients in winds, temperature, and moisture**
- ☞ **Depth is ~100 m A.G.L**
- ☞ **Layer where momentum , heat and moisture are exchanged between land and air by small turbulent eddies**

ABL roughly divide into three regions:

2. Mixed layer

The layer located above the surface layer (100- 1000 m) and below the entrainment zone.

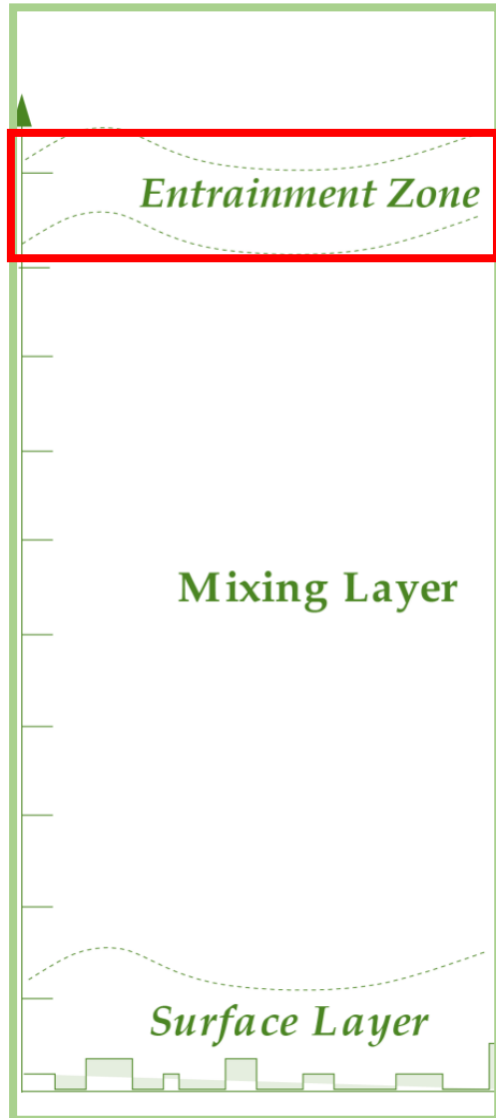


ABL roughly divide into three regions:

3. Entrainment Zone

A layer where the free atmosphere is entrainment into the top of the ML as a result of overshooting thermals

EZ is a transitional layer between ML and Free atmosphere- inversion layer (stable layer)

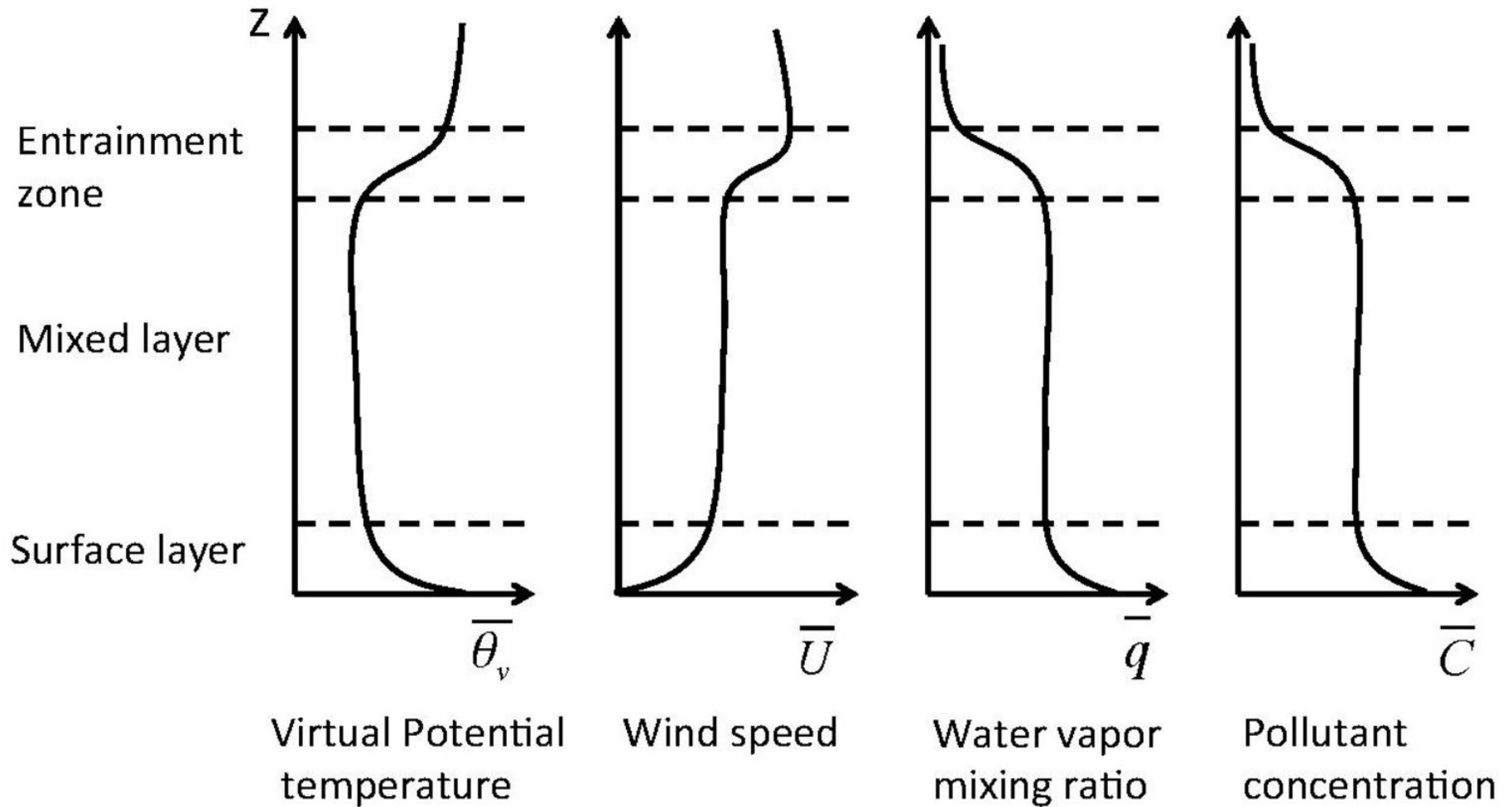


Located above the mixed-layer

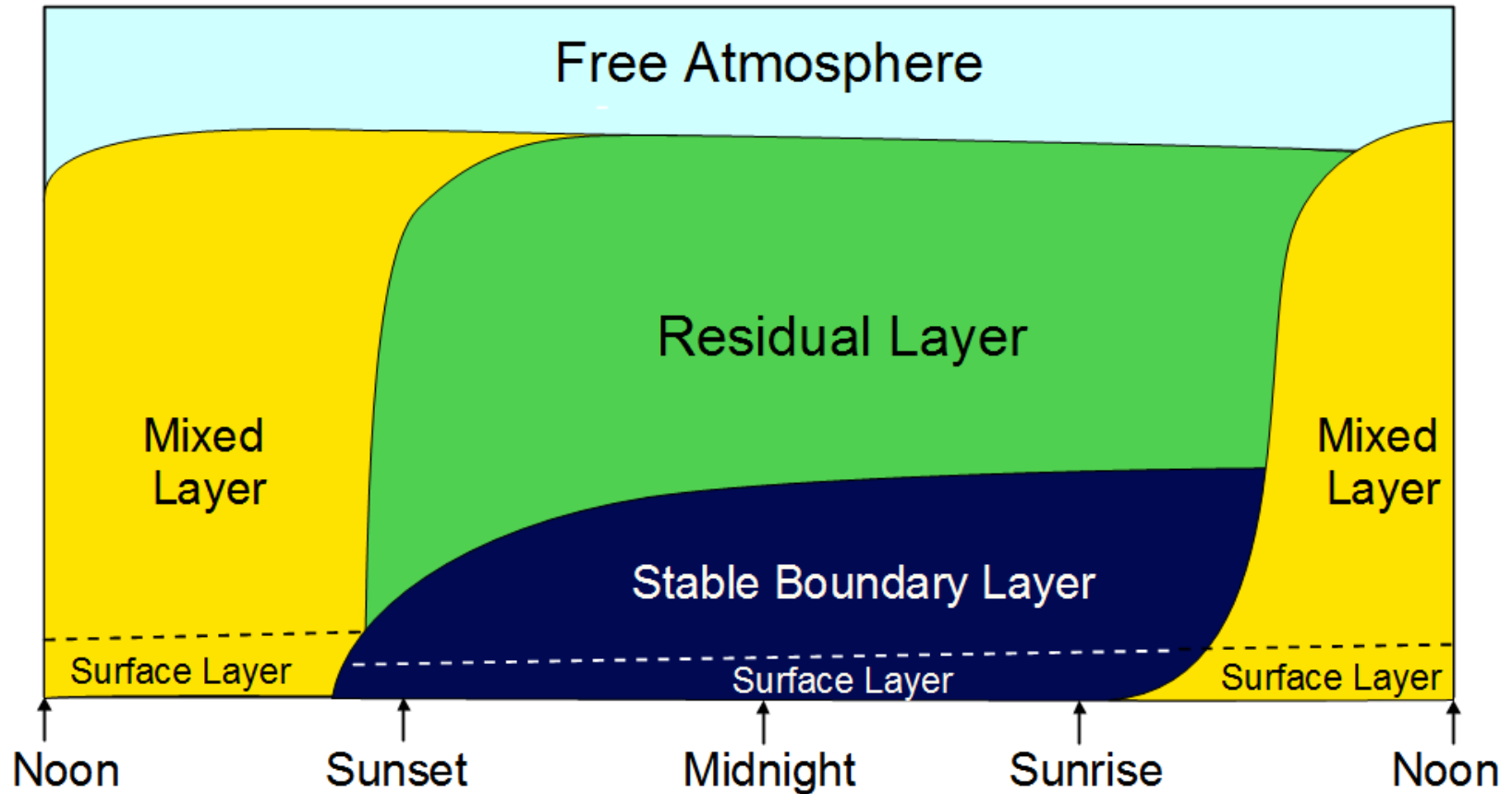
Characterized by:

- ☞ **Strong vertical gradients in winds, temperature, and moisture**
- ☞ **Depth is ~100 - 200 m A.G.L**
- ☞ **Often contains a temperature inversion, which suppress the turbulence down**

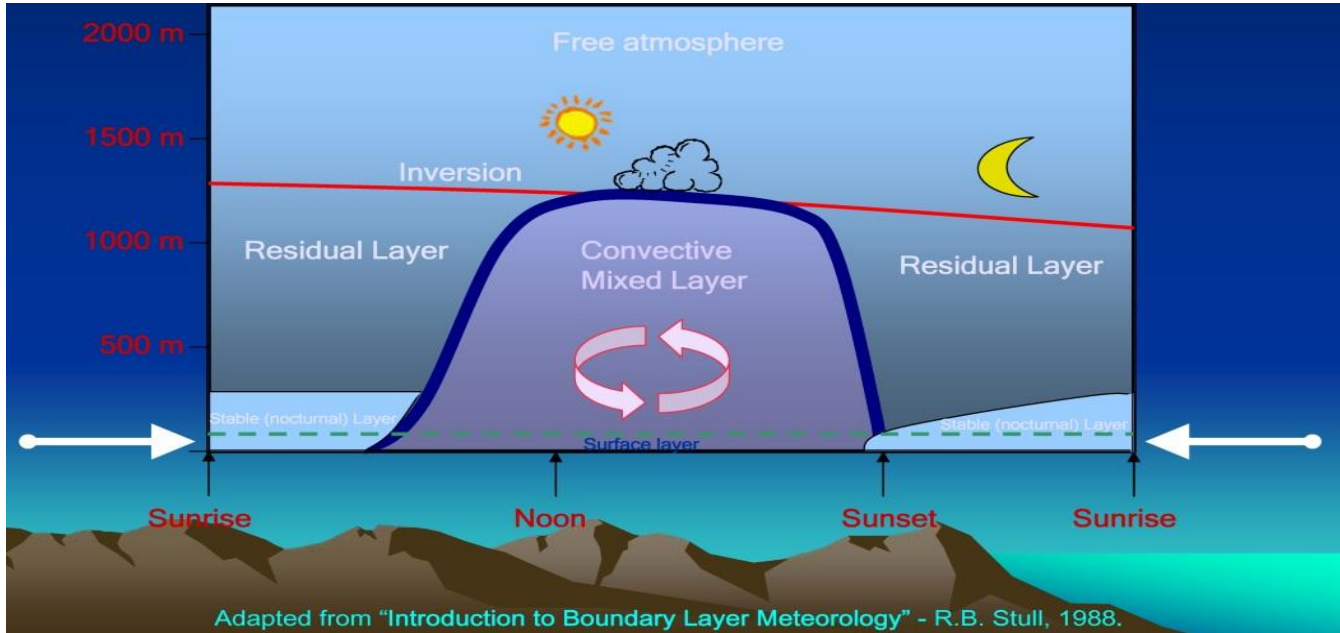
Vertical profile of mean variables in ABL:



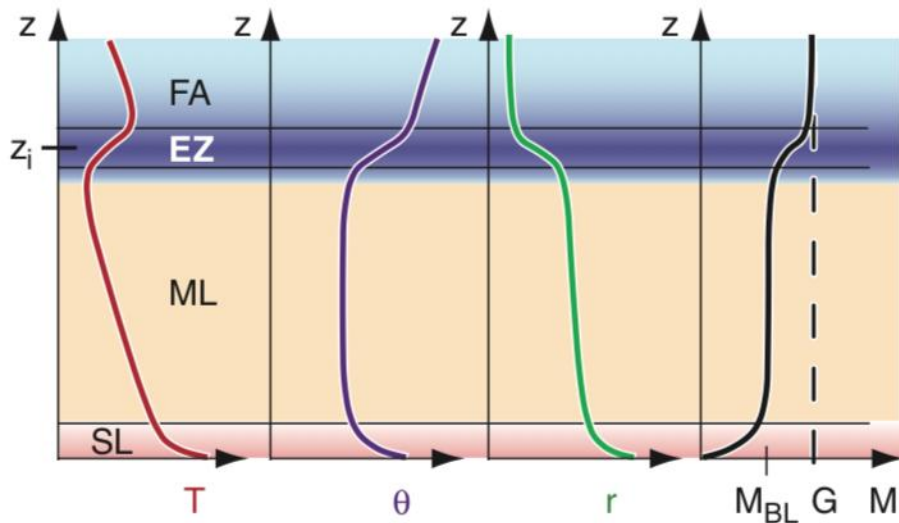
Daytime vs. Nighttime Turbulence



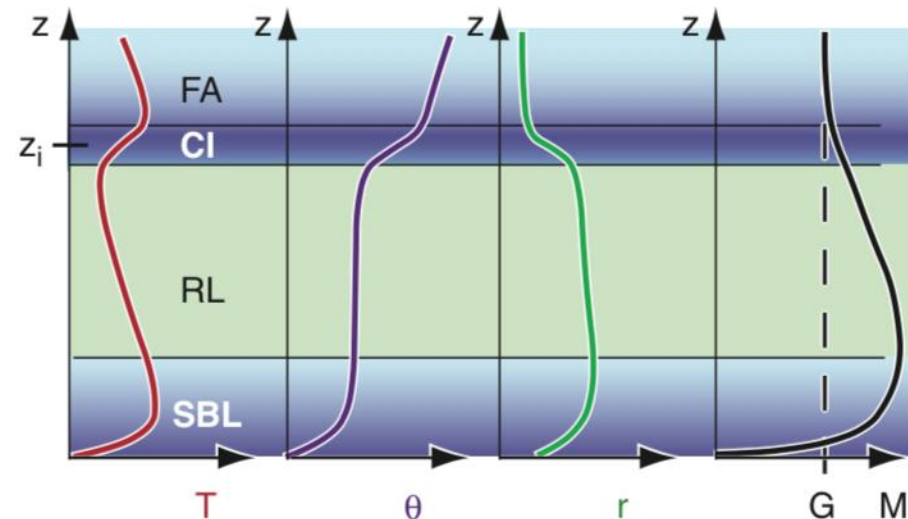
Evolution of the ABL



DAY



NIGHT

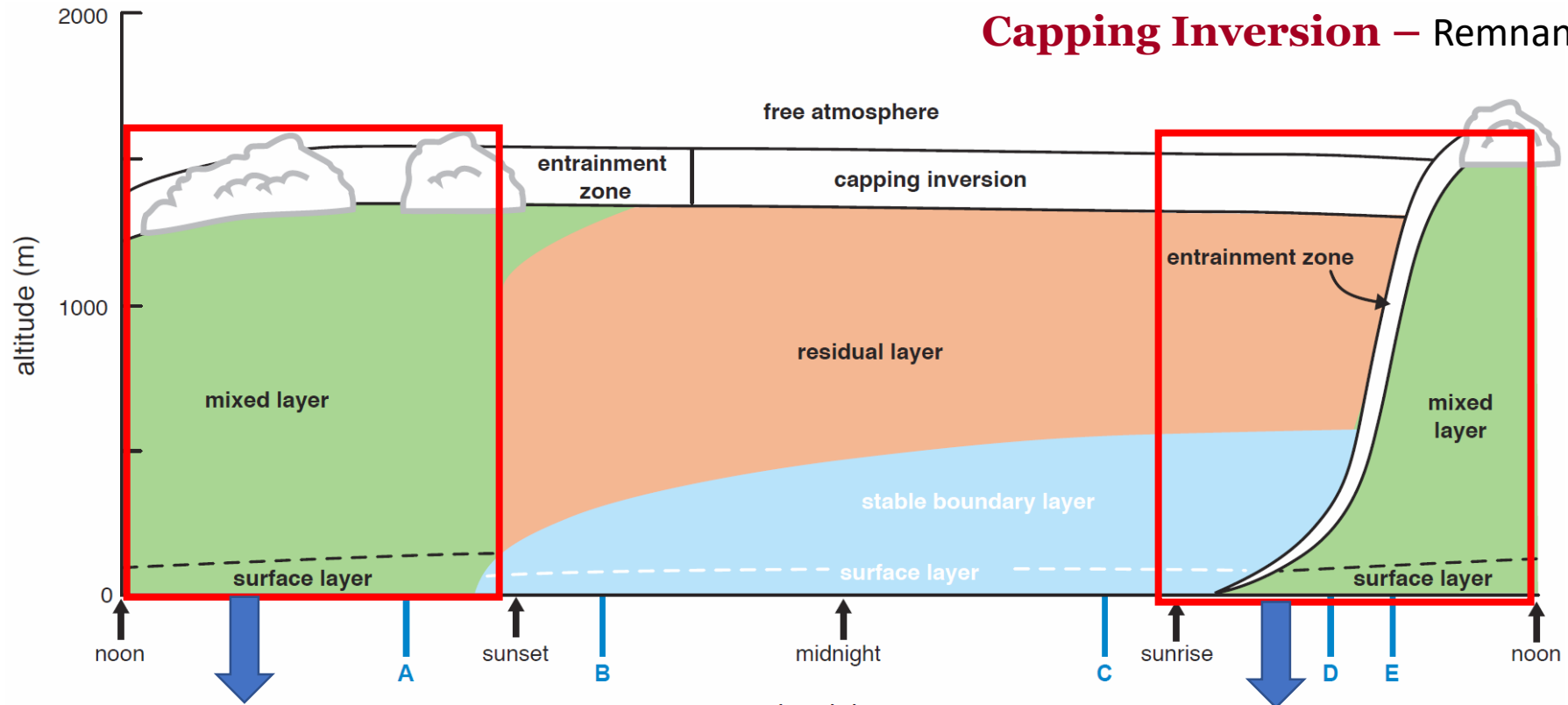


Diurnal Evolution of the ABL

Stable Layer

Residual Layer - Remnant ML

Capping Inversion – Remnant EZ



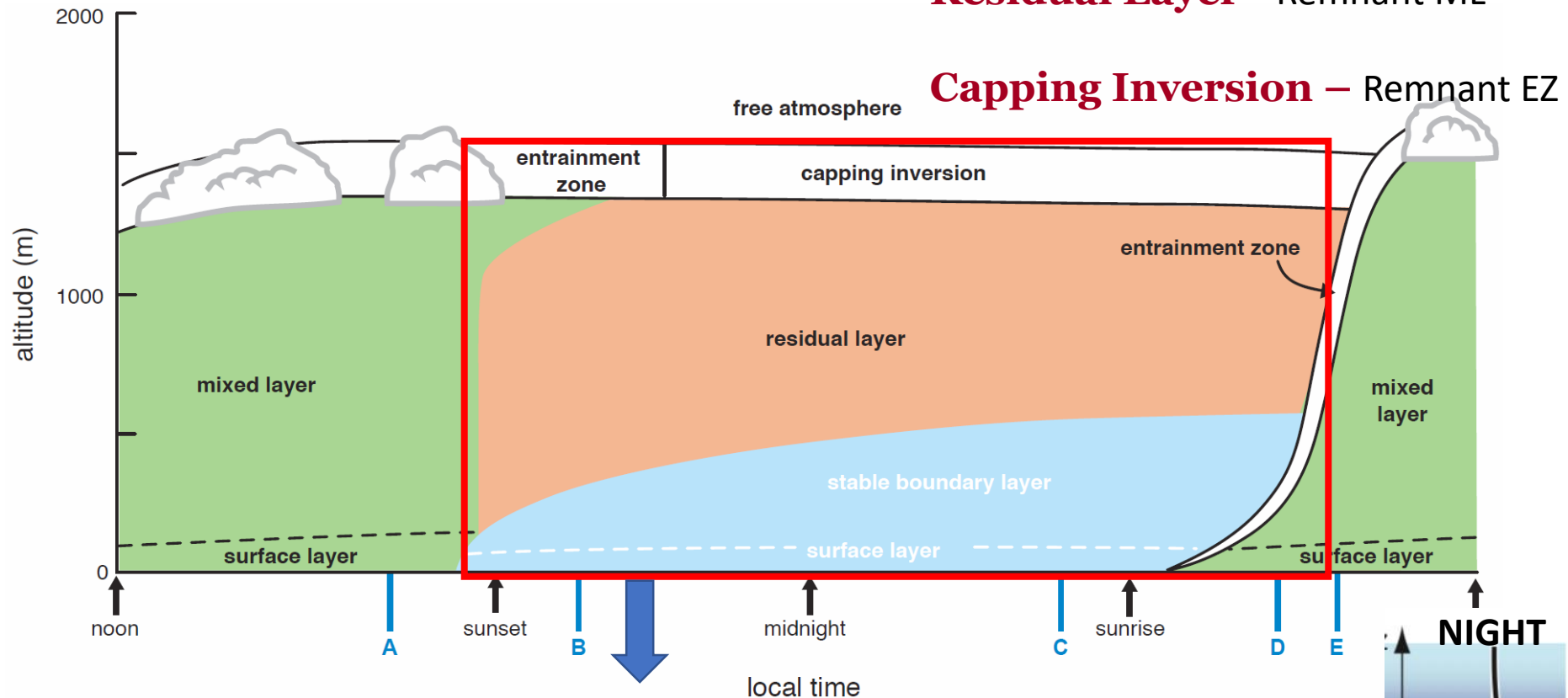
- After sunrise, surface heating (due to solar heating) causes turbulent eddies to develop- intense mixing, produces a mixed layer whose depth grows to a maximum at noon.
- Intense ML cause entrainment at the top of the mixed layer (in the entrainment zone).

Diurnal Evolution of the ABL

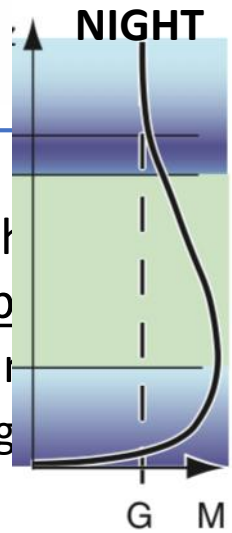
Stable Layer

Residual Layer - Remnant ML

Capping Inversion – Remnant EZ



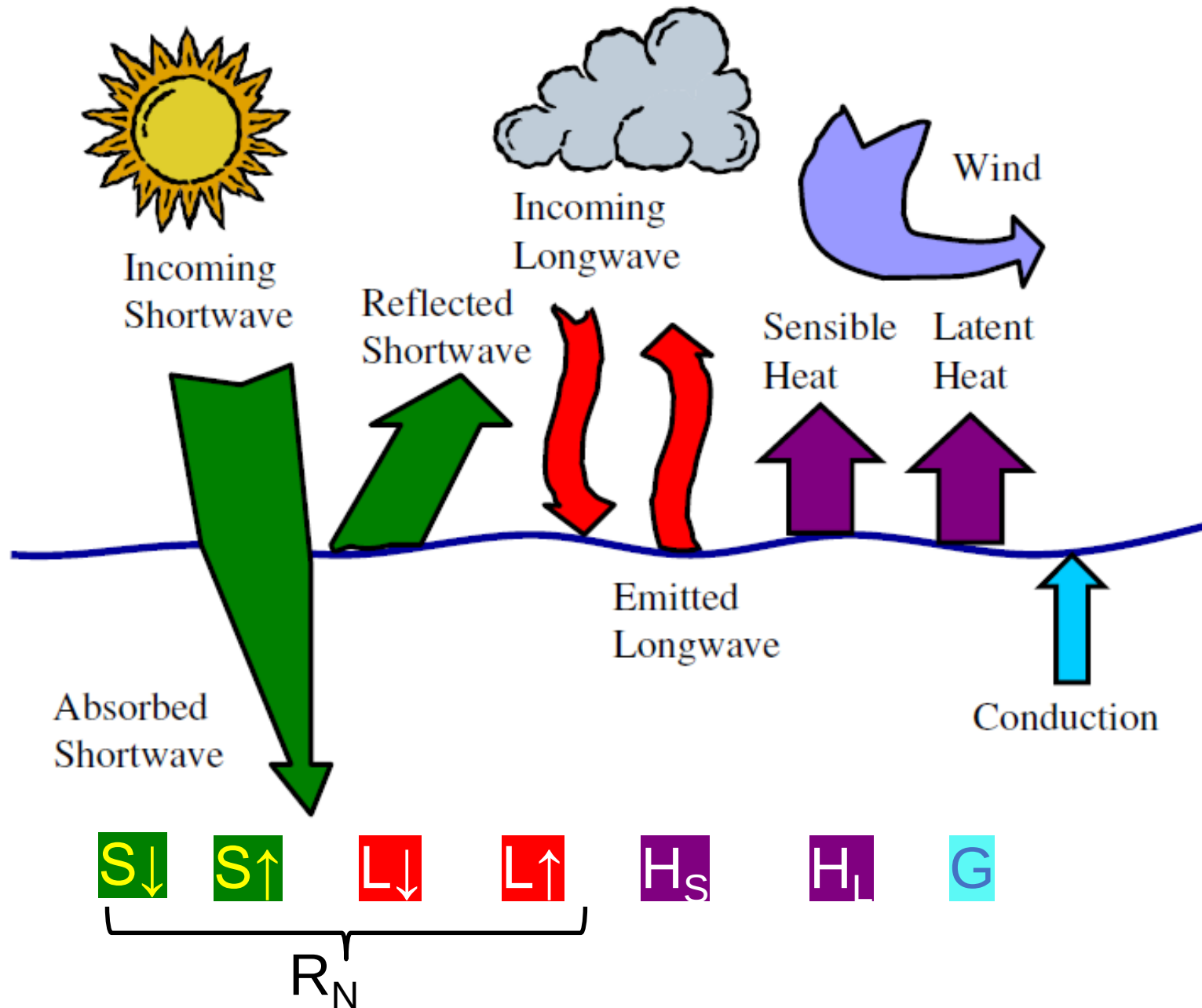
- At the sunset, longwave radiation cools the ground to temperatures, surface cooling --- > cease thermals, allowing turbulence to decay in the mixed layer. This former ML is now called the RL (basically the leftover p daytime ML) because it contains the residual moisture, heat, and pollutants.
- As night progresses, the residual layer is transformed by its contact with the ground into a stable (nocturnal) boundary layer.



Surface Energy Balance

- Surface Energy balance drives the diurnal variations in the ABL
- We have seen that boundary layer structure is strongly dependent on the strength of surface heating (unstable) or cooling (stable)
- Surface energy balance – the net sum of all energy fluxes “to and from” the surface

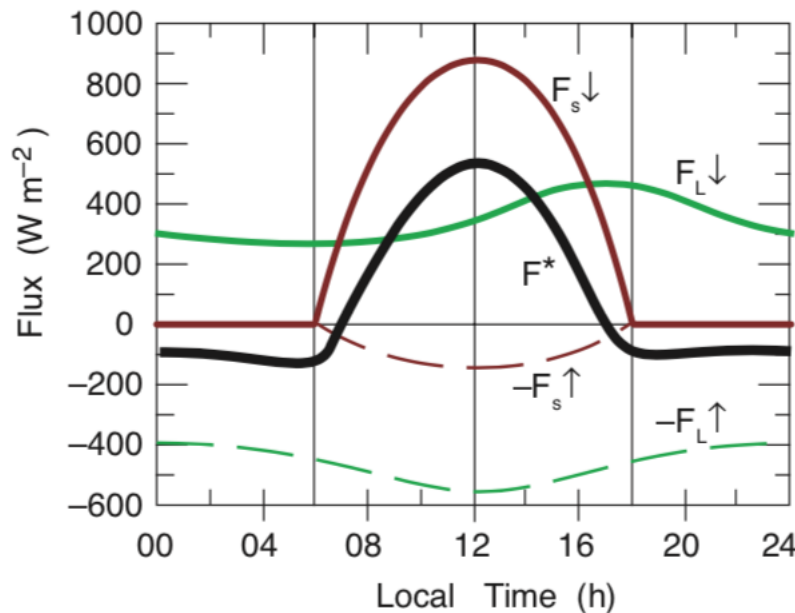
Surface Energy Balance



(A) Radiative Fluxes

The sum of the inputs to the surface minus the outputs yields the **net radiation flux F^*** absorbed at the surface:

$$F^* = F_s \downarrow - F_s \uparrow + F_L \downarrow - F_L \uparrow$$



Sign is defined such that positive implies an input to the surface

Sum of all these fluxes, F^* , is nearly constant and slightly negative during the night and Positive during daytime

(B) Turbulent Fluxes

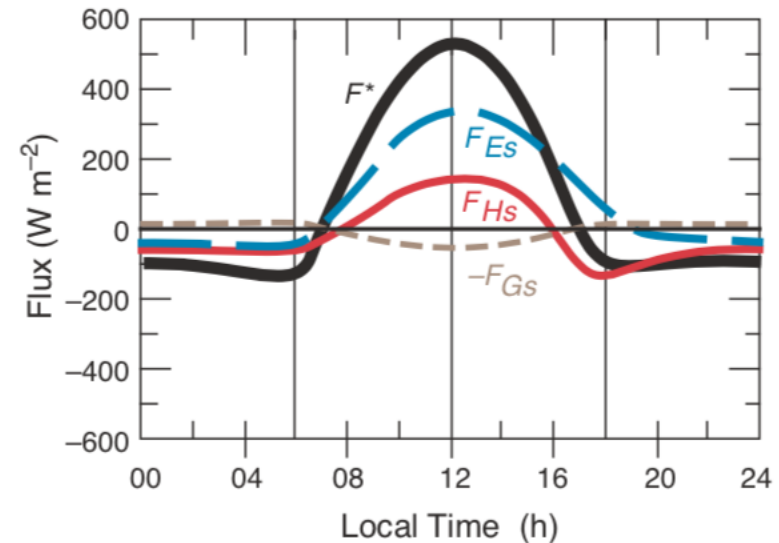
In addition to the radiative fluxes at the Earth's surface, the fluxes of sensible, latent heat and ground heat also need to be taken into account.

Sensible heat flux, F_{Hs} , into the air (positive upward, for flux away from the surface)

Latent heat flux, F_{Es} , into the air (positive upward),

Ground heat flux, F_{Gs} , down into the ground (positive downward, away from the surface)

$$F^* = F_{Hs} + F_{Es} + F_{Gs}$$



Heat Energy (Q) stored at the surface --- > Rfluxes- TFluxes

$$Q = F^* - F_{Hs} - F_{Es} - F_{Gs}$$

